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Math 441 — Midterm 1 practice

Total points: **100**. Please show your work and explain all answers. Calculators, computers, books and notes are **not** allowed. Suggestion: even if you cannot complete a problem, write out the part of the solution you know. You can get partial credit for it.

1. [**20 points**] Solve the following initial value problem for $y(x)$.

$$x\,y' - 2\,y = x^2$$
$$y(1) = 3$$

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2. [**20 points**] Consider the equation for $y(t)$:

$$\frac{\mathrm{d}y}{\mathrm{d}t} = y^2 - y - 2$$

Draw a sketch of the slope field (or draw several solution curves) and indicate on it the equilibrium solutions **and** their stability. What happens to the solution with initial condition $y(0) = 0$ as $t \to +\infty$? Verify your conclusion by calculating the analytic solution for this initial condition.

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3. [**20 points**] Find the solution for the following initial value problem for $y(x)$.

$$(\mathrm{e}^x \cos(y) + 2\,x\,y)\; y' + 2\,x + \mathrm{e}^x \sin(y) + y^2 = 0$$

$$y(1) = 0$$

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4. [**20 points**] An infinite tank is initially filled with 100 gallons of contaminated water which contains a concentration C_o of poison. Pure water (with no poison) is poured in the tank at a rate of 4 gallons per minute. At the same time water is let out of the tank at a rate of 2 gallons per minute. How long will it take for the water in the tank to have a concentration of poison that is one percent of the initial concentration?

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5. [**20 points**] Solve the initial value problem

$$y' = 2\,y + 1; \qquad y(0) = 0$$

using the method of successive approximations. Hint: start by rewriting it as an integral equation

$$y(x) = \int_0^x (2\,y(s) + 1)\,\mathrm{d}s$$

Check your answer by directly solving the initial value problem with another method of your choice.